MATHEMATICS

University *of* NORTH ALABAMA

Spring 2025 Math Problem of the Week: Problem #10

Due noon Monday, March 31
Open to all UNA undergraduates

Directions: Deliver your written solution to me (Dr. Bryant) in CPTMTH 328 by noon Monday, March 17 or, if you cannot find me, you may leave it with Ms. Buckley in CPTMTH 377. All work must be your own. You are required to justify your response fully. If you utilize a well-known theorem, state the theorem's full name and proposition, showing that the conditions of the theorem are met. Good luck!

Problem #10: Let $f : [a, b] \to \mathbb{R}$ be a continuous function that is twice differentiable in (a, b) and satisfies $f(a) = f(b) = 0$. Prove that

$$\int_a^b |f(x)|\, dx \leq \frac{1}{12} M(b-a)^3,$$

where $M = \sup |f''(x)|$ for $x \in (a, b)$.